

OPTIMIZING AIR TRAVEL

Flight Delay Analysis and Prediction

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Flight Delay Analysis: Project Overview

Objective:

To enhance the reliability and efficiency of air travel by analyzing historical flight data, uncovering delay patterns, and building predictive models for anticipating and mitigating flight delays using data-driven methodologies.

Key Goals:

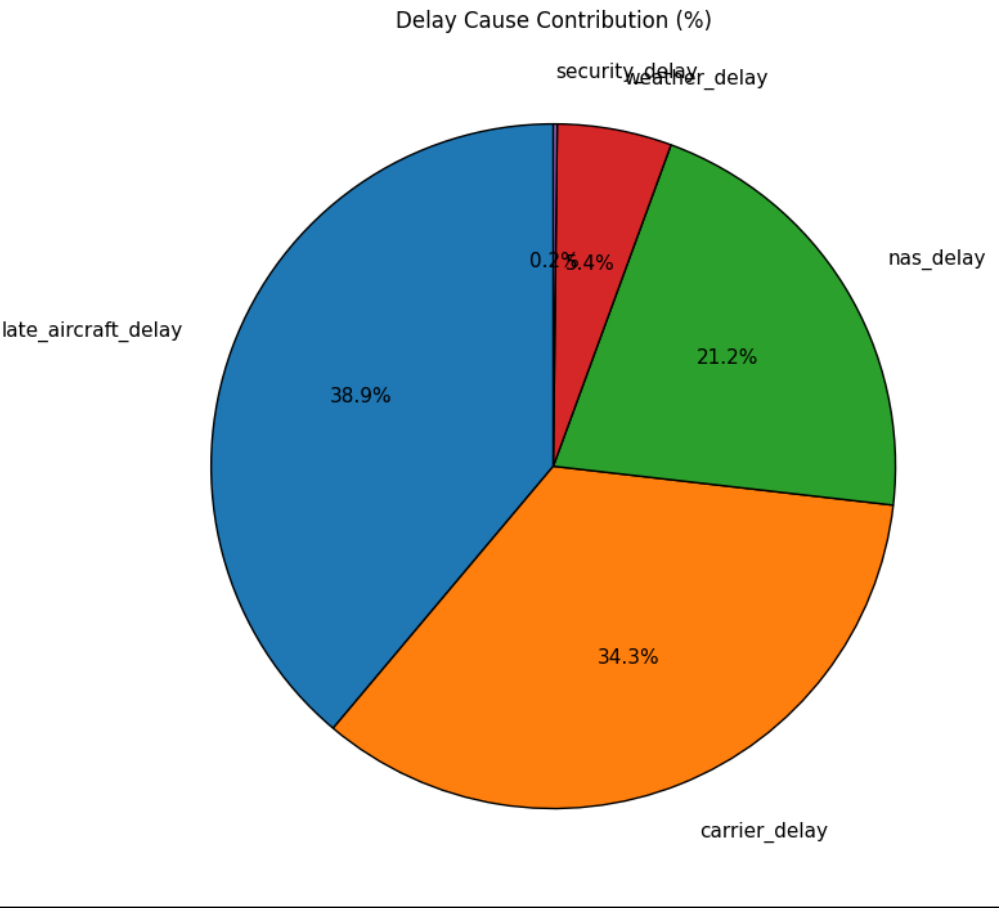
- Uncover Hidden Delay Patterns & Root Causes
- Develop Predictive Models
- Deliver Actionable Recommendations

Why Flight Delays Matter?

Flight delays disrupt travel, cause passenger frustration, and cost airlines billions annually.

Targets:

- Classification: Will a flight be delayed? (arr_del15)
- Regression: How long will the delay be? (arr_delay)

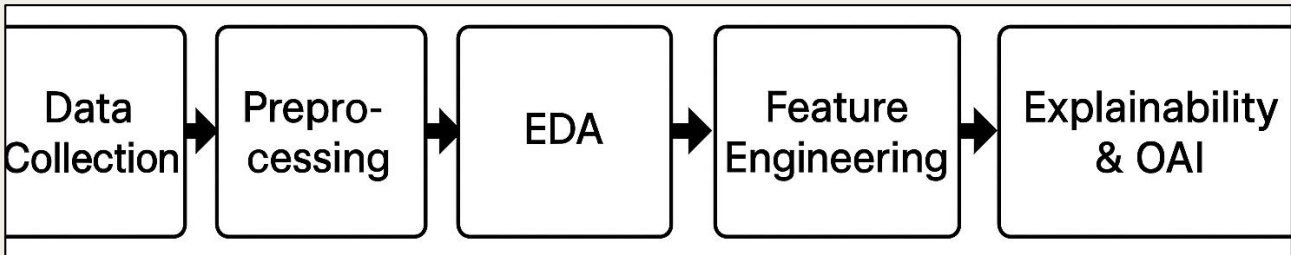


Top 3 Delay Causes

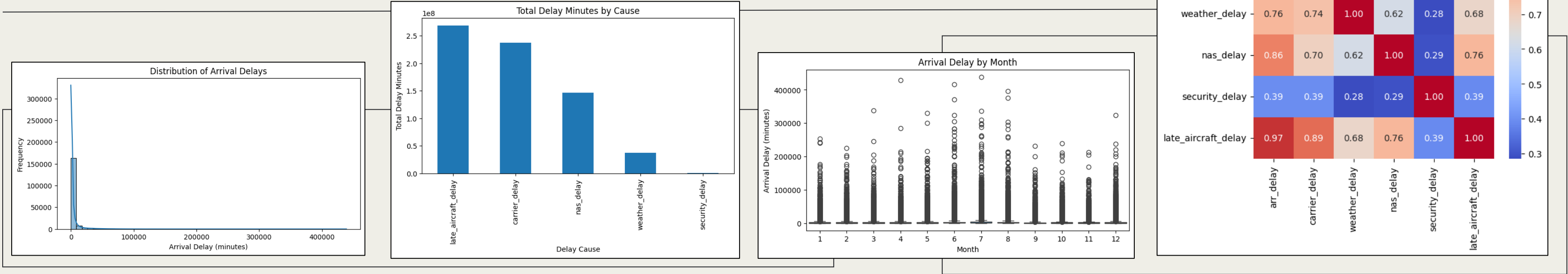
- Carrier Delay
Airline operational or crew-related issues
- Weather Delay
Adverse weather impacting flight schedules
- Late Aircraft Arrival
Previous flight of same aircraft was delayed

Approach (Methodology)

- Data Cleaning & Preprocessing
- Exploratory Data Analysis
- Feature Engineering
- Predictive Modeling
- Explainability (SHAP)
- Operational Adjustability Index (OAI)



Key Insights from Exploratory Data Analysis (EDA)



1. Distribution of Arrival Delays

- The **distribution is highly right-skewed**.
- Most flights are on time or experience only minor delays.
- A small proportion of flights suffer from extreme delays, creating a long tail in the data.

2. Common Causes of Delays

- **Top contributors** to total delay minutes are:
 - Late Aircraft Arrival (largest share)
 - Carrier-related Delays
 - NAS (National Aviation System) Delays
 - Weather-related Delays (significant but less than carrier/late aircraft)
- **Security delays** are relatively rare.

3. Temporal Patterns

- **Delays occur across all months**, but the presence of outliers in each month suggests periodic spikes (possibly related to weather, holidays, or operational bottlenecks).
- Consistent pattern of high outliers in summer and winter months may hint at seasonal effects.

4. Correlation Among Delay Variables

- **Strong correlations** between overall arrival delay and:
 - Carrier delay
 - Late aircraft delay
- Weather delays also show moderate correlation with overall delays.
- **Security delays** are less correlated, indicating they are usually isolated events.

Root Cause Analysis

1. Total Delay Minutes by Cause

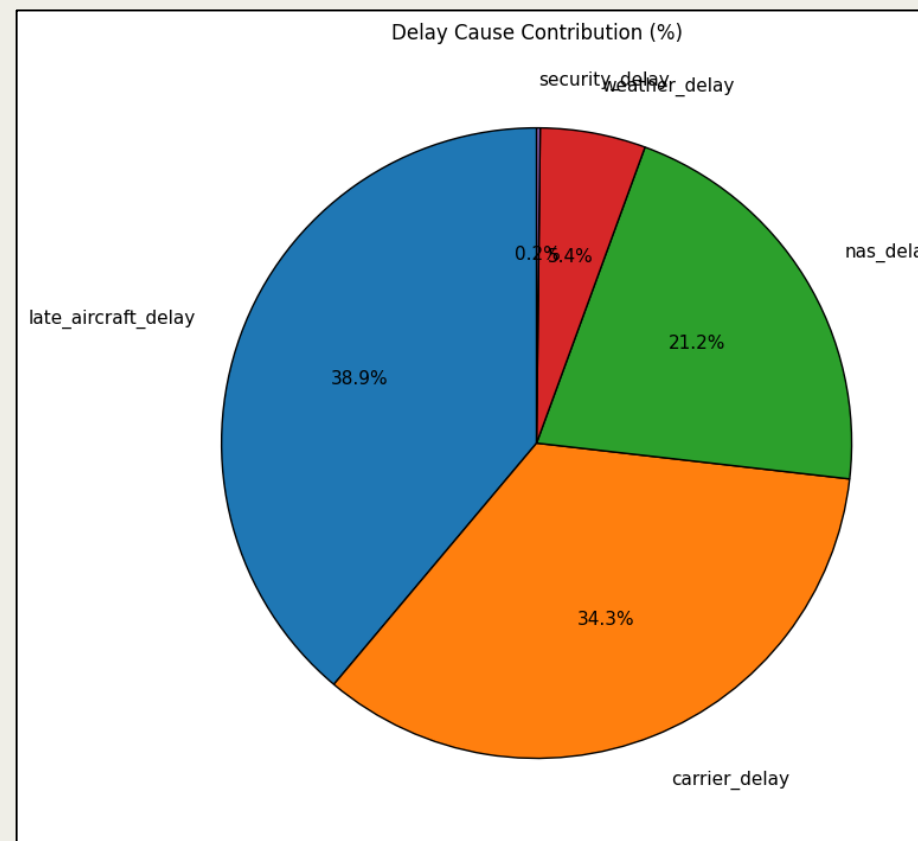
- **Late Aircraft Arrival** is the leading cause, contributing nearly **39%** of total delay minutes.
- **Carrier-related Delays** are the second highest at **34%**.
- **NAS (National Aviation System) Delays** account for **21%**.
- **Weather Delays** contribute about **5%**.
- **Security Delays** are negligible (<1%).

```
Total delay minutes by cause:  
late_aircraft_delay    268827517.0  
carrier_delay          237268793.0  
nas_delay              146768462.0  
weather_delay          37430676.0  
security_delay         1224329.0  
dtype: float64
```

```
Average delay minutes per record by cause:  
late_aircraft_delay    1501.854875  
carrier_delay          1325.546199  
nas_delay              819.949284  
weather_delay          209.113426  
security_delay         6.839941  
dtype: float64
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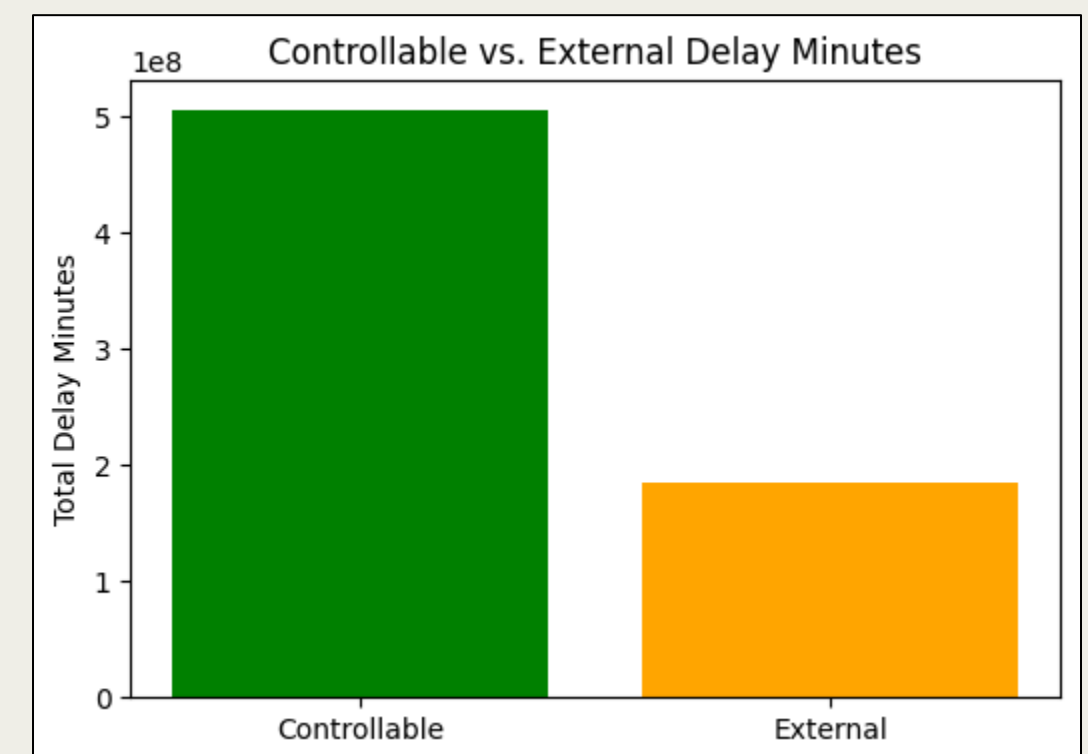
2. Average Delay Minutes per Flight by Cause

- Flights delayed due to **late aircraft** and **carrier issues** have the highest average delay per record.
- Weather and security delays are much lower on average.



3. Controllable vs. External Delays

- **Controllable Delays** (Carrier, Late Aircraft):
 - Together, these account for **over 70%** of all delay minutes.
 - This shows the **majority of delays are actionable** from an operational perspective.
- **External Delays** (Weather, NAS, Security):
 - Make up the remaining ~30%.



Predictive Modeling

1. Feature & Target Preparation

- Selected relevant features: flight timings, airline, airport codes, weather, and delay reasons.
- Defined targets:
 - **Classification:** Will the flight be delayed? (Yes/No)
 - **Regression:** If delayed, how many minutes?

2. Train-Test Split

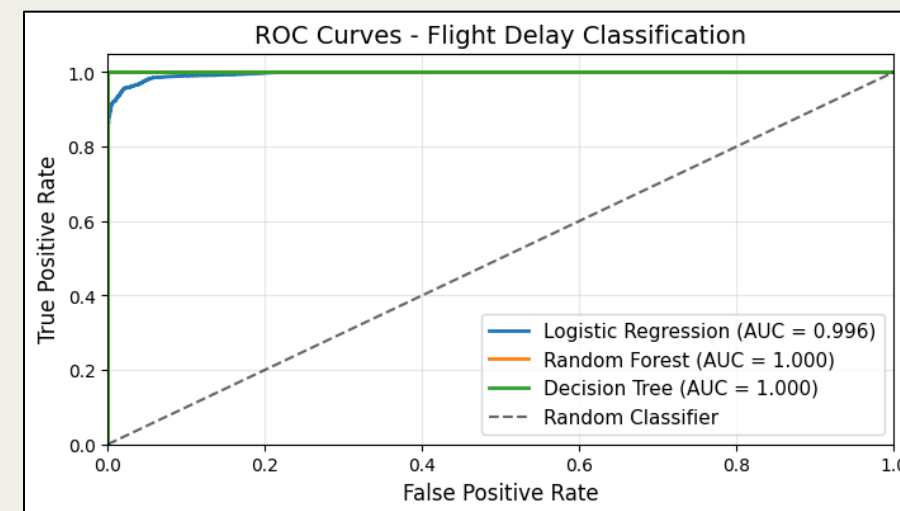
- Split the dataset into training and test sets to evaluate model performance fairly and avoid overfitting.

3. Data Scaling (for Linear Models)

- Applied feature scaling (standardization) to ensure fair comparison and optimal convergence in linear models.

4. Classification Models

- **Logistic Regression:**
 - Accuracy: 98.5%, ROC-AUC: 0.996
- **Random Forest & Decision Tree:**
 - Both achieved perfect accuracy (100%) and ROC-AUC (1.0) on test data.
- **Interpretation:**
 - Models reliably distinguish between delayed and on-time flights.

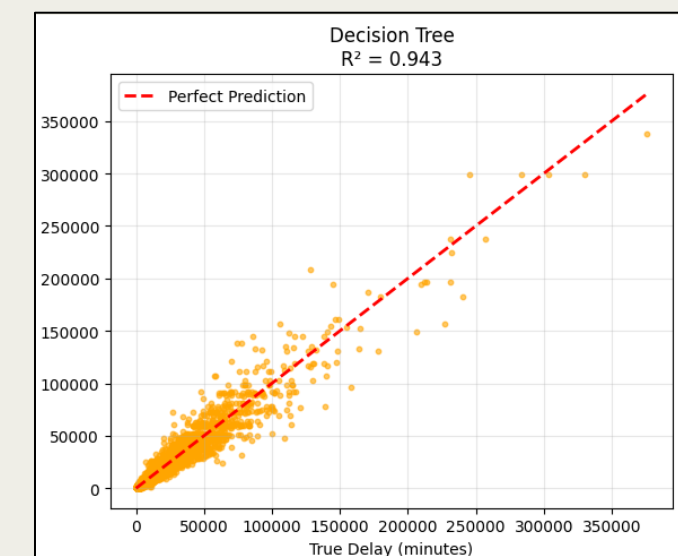
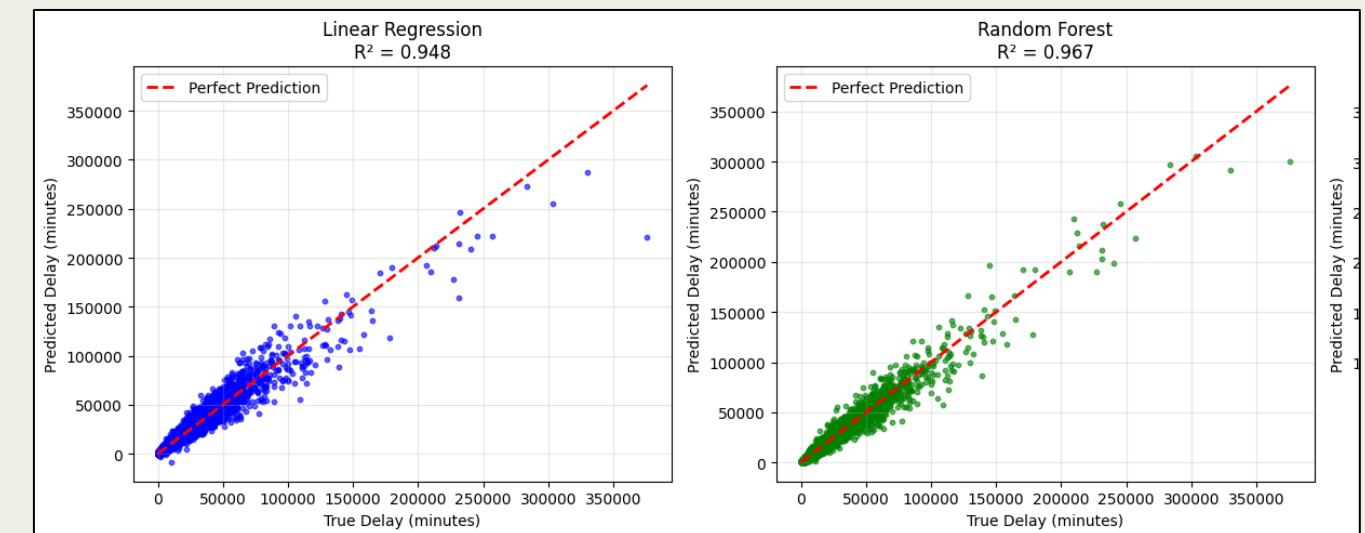


6. Operational Adjustability Index (OAI)

- Custom metric prioritizing delays airlines can control (e.g., carrier, late aircraft).
- Ensures the models are optimized for actionable, high-impact delay reduction.

5. Regression Models

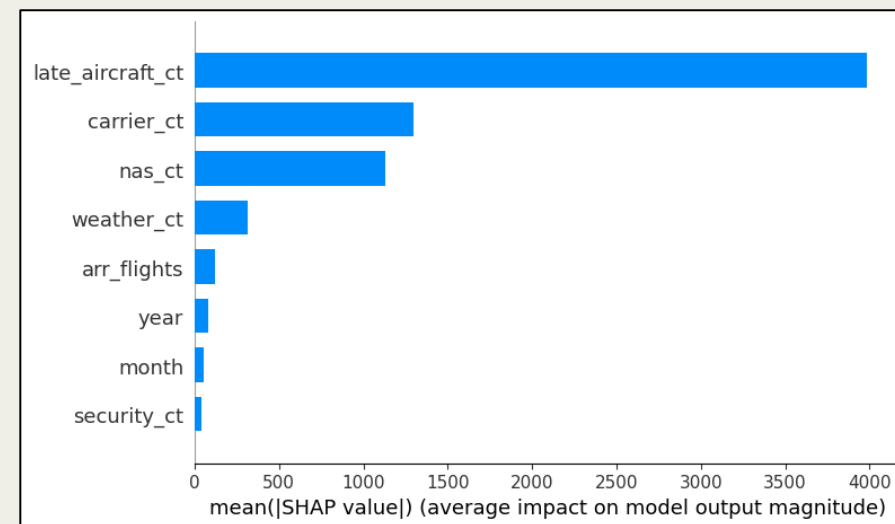
- **Linear Regression:**
 - MAE: 880.9 min, RMSE: 2583.7 min
- **Random Forest Regression:**
 - MAE: 650.7 min, RMSE: 2050.2 min (Best performer)
- **Decision Tree Regression:**
 - MAE: 801.4 min, RMSE: 2703.8 min
- **Interpretation:**
 - Ensemble methods outperform linear models in predicting delay duration.



Model Explainability with SHAP

Why Explainability?

Machine learning models can be complex “black boxes.” Using SHAP, we can interpret model predictions and understand which features drive flight delays.



Key SHAP Insights

- **Late Aircraft Count (late_aircraft_ct)** and **Carrier Delay Count (carrier_ct)** have the greatest impact on predicted delays, followed by NAS and Weather.
- Operational features (late aircraft, carrier) dominate model influence.

OAI-weighted SHAP: Focusing on What We Can Control

•Operational Adjustability Index (OAI):

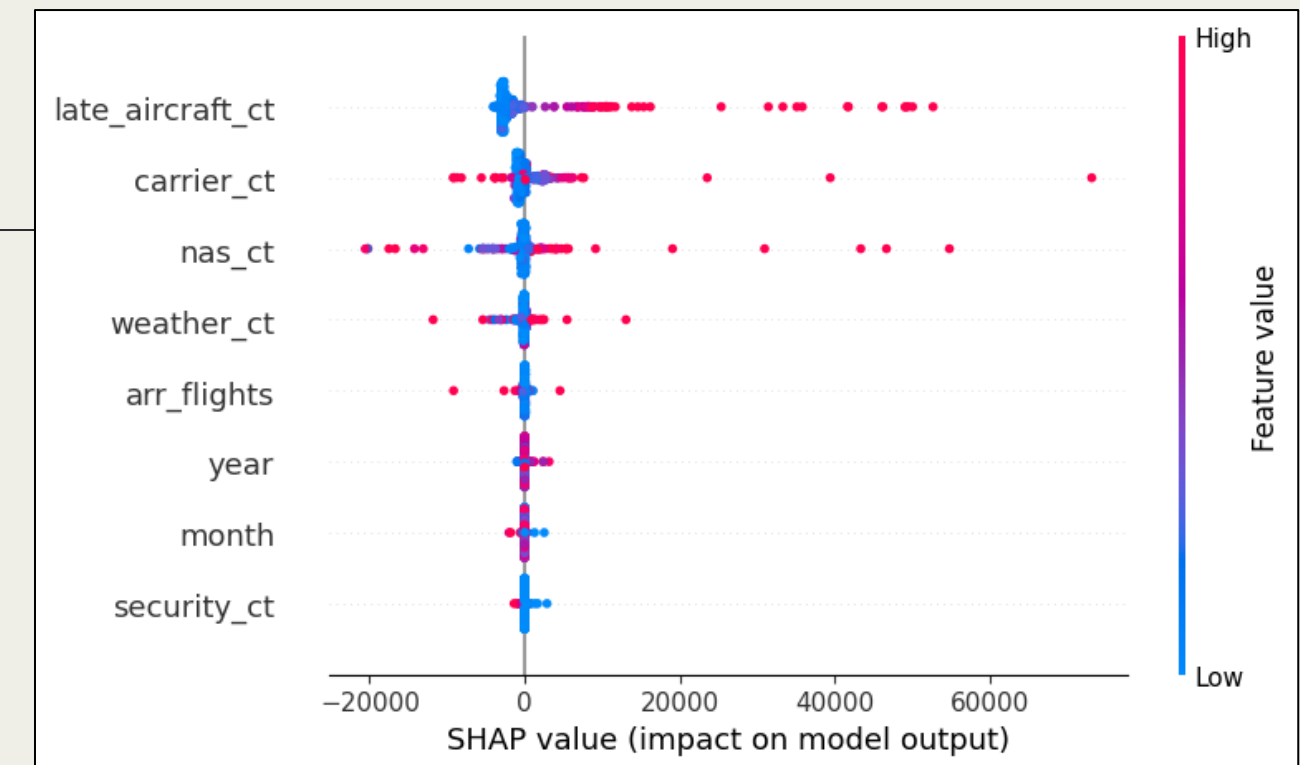
By assigning higher weights to controllable delay features, we measure which part of the model’s “explainability” is actionable.

•Result:

- **Controllable delay features** (carrier & late aircraft) contribute **over 90%** of the model’s explained variance.
- **External causes** (weather, NAS, security) contribute less than 7%.

•Strategic Takeaway:

Our models—and thus, our recommendations—are heavily focused on delays that airlines can actually influence.



This **SHAP Beeswarm plot** shows which features most influence the model’s predictions for flight delay duration. Each dot represents a flight, with its position indicating how much that feature increased or decreased the predicted delay. The plot reveals that late aircraft and carrier-related delays have the strongest impact on model output, especially when their values are high (shown in red). Other features, such as NAS and weather delays, have a smaller effect, while factors like month and security contribute minimally. Overall, the model relies most heavily on controllable operational delay counts to predict delay outcomes.

Actionable Recommendations

Adjust Flight Schedules

Analyze historical delay patterns by month and time of day.
Introduce optimal slack in schedules for flights and connections with recurrent delays, especially during high-risk periods.
Leverage predictive models to proactively identify high-risk flights and allocate buffer periods or re-time departures.

Improve Ground Operations & Resource Allocation

Focus delay reduction efforts on **carrier_delay** and **late_aircraft_delay**, which together account for **73.2%** of total delay minutes.
Audit turnaround processes, standardize ground operations, and invest in digital tools for crew coordination and efficiency.

Enhance Communication with Passengers

Implement real-time notifications for delays, gate changes, and updated ETAs.
Enable two-way messaging for easy rebooking, compensation, and customer support.

Prioritize Controllable Delay Factors

Deploy predictive analytics and SHAP explainability to pinpoint the flights and routes most affected by controllable delays.
Allocate extra resources and operational focus to these high-impact areas.

Collaborate on External Delay Reduction

While **26.8%** of delays are due to external factors (weather, NAS, security), airlines should **collaborate with airports and authorities** to share real-time data and develop joint contingency plans for smoother disruption management.

APPENDIX

1. Data Dictionary

Column Name	Column Definition
year	YYYY format
month	MM format (1-12)
carrier	US DOT code for unique airline carrier
carrier_name	Full airline name
airport	Official three-letter airport code
airport_name	Full airport name
arr_flights	Arrival flights
arr_del15	Arrival Delay Indicator (15+ min late = delayed; <15 min = on time)
carrier_ct	Carrier delay count (number of flights delayed by carrier issues)
weather_ct	Weather delay count
nas_ct	NAS (National Air System) delay count
security_ct	Security delay count
late_aircraft_ct	Late aircraft delay count
arr_cancelled	Number of flights cancelled
arr_diverted	Number of flights diverted
arr_delay	Arrival delay in minutes (actual - scheduled; can be negative for early arrivals)
carrier_delay	Carrier delay duration in minutes
weather_delay	Weather delay duration in minutes
nas_delay	NAS delay duration in minutes
security_delay	Security delay duration in minutes
late_aircraft_delay	Late aircraft delay duration in minutes

2. Methodological Details

Data Cleaning:

- Rows with missing arr_delay dropped.
- Numerical features imputed with median; categorical with mode.
- Feature engineering: Added season based on month.
- Categorical encoding: One-hot for carrier, airport, and season.

Modeling:

- Classification: Logistic Regression, Random Forest, Decision Tree; target: arr_del15 (delayed or not).
- Regression: Linear, Random Forest, Decision Tree; target: arr_delay (delay in minutes).
- Custom Metric: Operational Adjustability Index (OAI) — prioritizes controllable delays (carrier, late aircraft).
- Explainability: SHAP (SHapley Additive exPlanations) used to interpret feature impact, weighted by OAI.

Evaluation Metrics:

- Classification: Accuracy, Precision, Recall, F1-Score, ROC-AUC, Confusion Matrix.
- Regression: MAE, RMSE, R², Predicted vs. Actual scatter plots.
- Explainability: SHAP summary plots, OAI-weighted feature importance.

Thank you!
